2.1.5 Redox

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Oxidation number		
(a)	rules for assigning and calculating oxidation number for atoms in elements, compounds and ions	Learners will be expected to know oxidation numbers of O in peroxides and H in metal hydrides.
(b)	writing formulae using oxidation numbers	HSW8 Appropriate use of oxidation numbers in written communication.
(c)	use of a Roman numeral to indicate the magnitude of the oxidation number when an element may have compounds/ions with different oxidation numbers	Examples should include, but not be limited to, iron(II) and iron(III). Learners will be expected to write formulae from names such as chlorate(I) and chlorate(III) and *vice versa*. Note that 'nitrate' and 'sulfate', with no shown oxidation number, are assumed to be NO_3^- and SO_4^{2-}. HSW8 Systematic and unambiguous nomenclature.
Redox reactions		
(d)	oxidation and reduction in terms of: **(i)** electron transfer (ii) changes in oxidation number	Should include examples of s-, p- and d-block elements.
(e)	redox reactions of metals with acids to form salts, including full equations **(see also 2.1.4 c)**	Metals should be from s-, p- and d-blocks e.g. Mg, A*l*, Fe, Zn. Ionic equations **not** required. In **(e)**, reactions with acids will be limited to those producing a salt and hydrogen. Reactions involving nitric acid or concentrated sulfuric acid could be assessed in the context of **(f)**.
(f)	interpretation of redox equations in **(e)**, and unfamiliar redox reactions, to make predictions in terms of oxidation numbers and electron loss/gain.	*M0.2*